

HIMANSHU GUPTA

Data Scientist | NLP | Machine Learning

[Portfolio](#) • [LinkedIn](#) • [GitHub](#)

PROFESSIONAL SUMMARY

AI & Machine Learning undergraduate specializing in Natural Language Processing and applied Data Science. Built a next-word prediction language model with PyTorch and LSTMs, an NLP spam classifier (98.84% accuracy, 95.24% F1-score), and a customer satisfaction prediction system (79.2% accuracy, XGBoost + SHAP). Skilled across supervised/unsupervised ML, deep learning, and NLP (TF-IDF, Word2Vec, n-grams). Seeking a Data Scientist or NLP-focused fresher role/internship.

CORE SKILLS

- Programming & Databases: Python, SQL
- NLP: Text Preprocessing (Tokenization, Stopword Removal, Stemming), Bag-of-Words, N-Grams, TF-IDF, Word2Vec, NLTK
- Machine Learning: Supervised & Unsupervised Learning, Regression, Classification, Naive Bayes, Ensemble Learning, Feature Engineering, Hyperparameter Tuning (GridSearchCV), Model Evaluation, SMOTE, SHAP
- Deep Learning: ANNs, CNNs, RNNs/LSTMs, Backpropagation, Transfer Learning (MobileNetV2)
- Frameworks & Libraries: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, Pandas, NumPy
- Data Analysis & Visualization: Excel, Power BI, DAX
- Tools & Deployment: Git, GitHub, Streamlit

PROJECTS

Gutenberg Next-Word Predictor

Deep Learning / NLP

Python, PyTorch, LSTM, NLP | [GitHub](#)

- Built a word-level next-word prediction language model in PyTorch, trained on cleaned Project Gutenberg text using an Embedding layer followed by stacked LSTM layers.
- Designed the full pipeline — text cleaning, tokenization, and fixed-length context windows — training with cross-entropy loss and CUDA GPU acceleration.
- Applied early stopping based on validation loss to manage overfitting, reaching 15.86% top-1 and 33.99% top-5 validation accuracy.
- Implemented top-k sampling for text generation, producing coherent short continuations from arbitrary prompts, and saved model/tokenizer checkpoints for reuse.

Email/SMS Spam Classifier

NLP

Python, NLTK, Scikit-learn, TF-IDF, Naive Bayes, Streamlit | [GitHub](#) | [Streamlit App](#)

- Built an end-to-end NLP pipeline to classify messages as spam or ham, including tokenization, stopwords removal, and stemming for text preprocessing.
- Compared CountVectorizer and TF-IDF feature extraction across five models (Decision Tree, Random Forest, Gaussian/Bernoulli/Multinomial Naive Bayes), evaluating each on accuracy, precision, recall, and F1-score.
- Selected TF-IDF + Bernoulli Naive Bayes as the final model, achieving 98.84% accuracy and a 95.24% F1-score, and deployed it as an interactive Streamlit web application.

Customer Satisfaction (CSAT) Prediction

Python, Scikit-learn, XGBoost, SHAP, Streamlit | [GitHub](#) | [Streamlit App](#)

- Built an end-to-end ML pipeline combining structured and textual features to flag dissatisfied customers before escalation.
- Benchmarked Logistic Regression, Random Forest, Decision Tree, and XGBoost with SMOTE and GridSearchCV tuning; XGBoost delivered the best precision/recall balance (0.804 F1, 79.2% accuracy).
- Applied SHAP for model interpretability and deployed a Streamlit app for real-time prediction and feature-impact scenario testing.

EDUCATION

B.Tech, Artificial Intelligence & Machine Learning

Vivekananda Institute of Professional Studies, New Delhi | 2023 – 2027